

Lecture 11: 02/09/2026

Lecturer: Mauricio Daros

11.1 Continuous Time Markov Chains on Discrete Spaces

A continuous-time markov chain is like a discrete-time markov chain, but on continuous time.

11.1.1 Introduction

Let $X = \{X(t) : t \geq 0\}$ be a family of random variables taking values in a countable state space S and indexed by the half-line $[0, \infty)$. As before, we shall assume that S is a subset of the integers. The process X is called a *continuous-time Markov chain* if it satisfies the following condition.

Definition 11.1.1. The process X satisfies the *Markov property* if

$$\mathbb{P}(X(t_n) = j \mid X(t_1) = i_1, \dots, X(t_{n-1}) = i_{n-1}) = \mathbb{P}(X(t_n) = j \mid X(t_{n-1}) = i_{n-1}) \quad (11.1)$$

for all $j, i_1, \dots, i_{n-1} \in S$ and every sequence of times $t_1 < t_2 < \dots < t_n$.

Remark 11.1.1. Let $\mathcal{F}_s = \sigma(X(u) : u \leq s)$. Then (11.1) is equivalent to

$$\mathbb{P}(X(t) = j \mid \mathcal{F}_s) = \mathbb{P}(X(t) = j \mid X(s)),$$

for every $j \in S$ and $s \leq t$.

Definition 11.1.2. The process X is *time-homogeneous* if, for every $i, j \in S$ and $s, t \geq 0$,

$$\mathbb{P}(X(s+t) = j \mid X(s) = i) = \mathbb{P}(X(t) = j \mid X(0) = i).$$

Henceforth, we suppose that X is a time-homogeneous Markov chain on a countable space S . Define

$$p_{ij}(t) = \mathbb{P}(X(t) = j \mid X(0) = i),$$

and let $\mathbf{P}_t$ be the $|S| \times |S|$ matrix with entries $p_{ij}(t)$.

11.1.2 Examples

11.1.2.1 Two state chain

Let $S = \{0, 1\}$, and fix rates $\lambda, \mu > 0$. Let $X(0)$ be an S -valued random variable, and define $Y_0 = X(0)$ and $Y_{n+1} = 1 - Y_n$ for $n \geq 0$. Thus, Y_n is the state occupied after n jumps. Conditional on $X(0)$, let $H_1, H_2, \dots$ be independent random variables such that

$$H_{n+1} \sim \begin{cases} \text{Exp}(\lambda), & Y_n = 0, \\ \text{Exp}(\mu), & Y_n = 1. \end{cases}$$

Set $T_0 = 0$, $T_n = H_1 + \dots + H_n$ for $n \geq 1$, and define

$$X(t) = Y_n \quad \text{when } T_n \leq t < T_{n+1}.$$

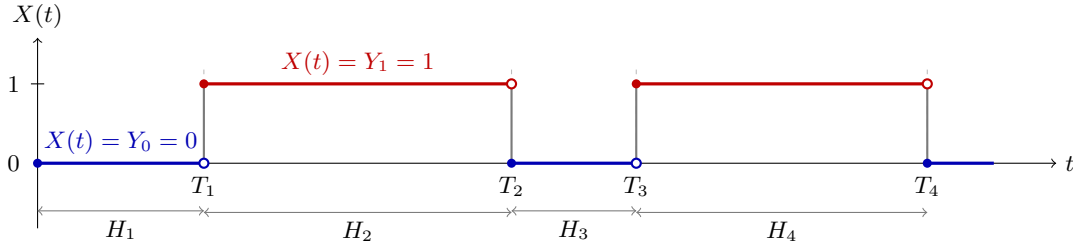


Figure 11.1: A sample path. On $[T_n, T_{n+1})$, the path has value $X(t) = Y_n$. Each holding-time interval has length $H_{n+1} = T_{n+1} - T_n$ (and $T_n = H_1 + \dots + H_n$).

In other words, the chain alternates between 0 and 1, remaining in state 0 for an exponential time of rate λ and in state 1 for an exponential time of rate μ .

To see that X is Markovian, fix $t_1 < \dots < t_n$ and $j, i_1, \dots, i_{n-1} \in S$. Given that $X(t_{n-1}) = i_{n-1}$, the residual time until the next jump is exponential with rate λ if $i_{n-1} = 0$ and with rate μ if $i_{n-1} = 1$, by the memoryless property of the exponential distribution. The holding times after that jump are independent of the states visited before time t_{n-1} . Consequently, (11.1) follows (how would you show this paragraph with equations?).

We now show an inefficient way of computing the transition probabilities by considering all possible numbers of jumps. Later on we will see how to do it more efficiently. Write $\mathbb{P}_i(\cdot) = \mathbb{P}(\cdot \mid X(0) = i)$. Since the chain alternates between the two states, starting from 0 it is in state 0 after an even number of jumps and in state 1 after an odd number of jumps. Thus,

$$\begin{aligned} p_{00}(t) &= \sum_{m=0}^{\infty} \mathbb{P}_0(T_{2m} \leq t < T_{2m+1}), & p_{01}(t) &= \sum_{m=0}^{\infty} \mathbb{P}_0(T_{2m+1} \leq t < T_{2m+2}), \\ p_{10}(t) &= \sum_{m=0}^{\infty} \mathbb{P}_1(T_{2m+1} \leq t < T_{2m+2}), & p_{11}(t) &= \sum_{m=0}^{\infty} \mathbb{P}_1(T_{2m} \leq t < T_{2m+1}). \end{aligned}$$

For example, the first two paths contributing to $p_{00}(t)$ are the path with no jumps, of probability $\mathbb{P}_0(H_1 > t)$, and the path $0 \rightarrow 1 \rightarrow 0$, of probability $\mathbb{P}_0(H_1 + H_2 \leq t < H_1 + H_2 + H_3)$. Each term is computed from the independent exponential holding times H_n ; evaluating the four sums gives the transition matrix below.

The transition matrix is given by

$$\mathbf{P}_t = \frac{1}{\lambda + \mu} \begin{pmatrix} \mu + \lambda e^{-(\lambda + \mu)t} & \lambda(1 - e^{-(\lambda + \mu)t}) \\ \mu(1 - e^{-(\lambda + \mu)t}) & \lambda + \mu e^{-(\lambda + \mu)t} \end{pmatrix}.$$

For instance, $p_{01}(t)$ is the probability that a chain starting in state 0 is in state 1 at time t .

11.1.2.2 Poisson Process

Fix $\lambda > 0$, and let $\tau_1, \tau_2, \dots$ be iid $\text{Exp}(\lambda)$ random variables. Set $T_0 = 0$, $T_k = \sum_{j=1}^k \tau_j$ for $k \geq 1$, and

$$X_t = \sum_{k=1}^{\infty} \mathbf{1}\{T_k \leq t\}, \quad t \geq 0.$$

The strong law of large numbers gives $T_k/k \rightarrow 1/\lambda$ almost surely, so there are only finitely many arrivals on each bounded time interval. Thus X has càdlàg paths with jumps of size one. We can think of the T_k as the arrival times of telephone calls at a call center; then X_t counts the calls received by time t . The process $(X_t)_{t \geq 0}$ is called a *Poisson process of rate λ* .

Theorem 11.1.1. *The Poisson process $(X_t)_{t \geq 0}$ defined above is a Markov chain taking values in $\mathbb{N}$. Moreover it satisfies*

(i) $X_0 = 0$ almost surely;

(ii) increments are independent: for every $0 \leq t_0 < \dots < t_m$, the random variables $(X_{t_j} - X_{t_{j-1}})_{j=1}^m$ are independent.

(iii) $X_t - X_s \sim \text{Poisson}(\lambda(t-s))$ for every $0 \leq s < t$.

Furthermore, any càdlàg process satisfying (i), (ii) and (iii) is a Poisson process.

Proof. We first prove the Markov property using memorylessness. Fix $s \geq 0$ and n with $\mathbb{P}(X_s = n) > 0$. On $\{X_s = n\}$, the arrivals observed so far are $T_1, \dots, T_n$, and the next waiting time satisfies $\tau_{n+1} > s - T_n$. Independence and memorylessness give, almost surely under this conditioning, for $u \geq 0$,

$$\mathbb{P}(T_{n+1} - s > u \mid T_1, \dots, T_n, X_s = n) = \frac{e^{-\lambda(s-T_n+u)}}{e^{-\lambda(s-T_n)}} = e^{-\lambda u}.$$

Moreover, $\tau_{n+2}, \tau_{n+3}, \dots$ are independent of $(\tau_1, \dots, \tau_{n+1})$, so they remain iid exponentials independent of the residual time after conditioning. Thus the waiting times after s are iid $\text{Exp}(\lambda)$, with a conditional law that does not depend on the observed arrival times.

On $\{X_s = n\}$, every earlier count is determined by $T_1, \dots, T_n$. Averaging over these arrival times therefore gives, for any $0 \leq s_1 < \dots < s_m = s$ and count history of positive probability with $n_m = n$,

$$\begin{aligned} \mathbb{P}(X_{s+h} = j \mid X_{s_1} = n_1, \dots, X_{s_m} = n_m) &= \mathbb{P}(X_h = j - n) \\ &= \mathbb{P}(X_{s+h} = j \mid X_s = n), \quad h \geq 0. \end{aligned}$$

The last equality is the same argument with only the observation at s . This proves the Markov property and time homogeneity.

We now verify (i)–(iii). Since $\tau_1 > 0$ almost surely, $X_0 = 0$, proving (i). The same argument shows that the conditional law of $X_{s+h} - X_s$ given any finite count history up to s is the law of X_h . Applying this successively at $t_0, \dots, t_{m-1}$ proves that the increments in (ii) are independent.

For (iii), put $q_k(h) = \mathbb{P}(X_h = k)$. First, $q_0(h) = \mathbb{P}(\tau_1 > h) = e^{-\lambda h}$. After the first arrival, the waiting times $\tau_2, \tau_3, \dots$ are independent of τ_1 and have the same joint law as the original sequence. Integrating over the density of τ_1 therefore gives, for $k \geq 1$,

$$q_k(h) = \int_0^h \lambda e^{-\lambda u} q_{k-1}(h-u) du.$$

Induction on k now yields, for $h > 0$,

$$q_k(h) = \frac{\lambda^k e^{-\lambda h}}{(k-1)!} \int_0^h (h-u)^{k-1} du = e^{-\lambda h} \frac{(\lambda h)^k}{k!}, \quad k \geq 1.$$

Thus $X_h \sim \text{Poisson}(\lambda h)$. Since $X_t - X_s$ has the same distribution as X_{t-s} , (iii) follows.

Conversely, let Z be a càdlàg process satisfying (i)–(iii). Its values at finitely many times are partial sums of the increments, so Z and X have the same finite-dimensional distributions. On rational times, Z is nondecreasing and non-negative integer-valued outside a single null set; right-continuity extends these properties to all times. Set

$$S_0 = 0, \quad S_k = \inf\{t \geq 0 : Z_t \geq k\}, \quad k \geq 1,$$

with $\inf \emptyset = \infty$. Monotonicity and right-continuity imply $\{S_k \leq t\} = \{Z_t \geq k\}$ up to a null set. Thus, for any $r \geq 1$ and $s_1, \dots, s_r \geq 0$,

$$\begin{aligned} \mathbb{P}(S_1 \leq s_1, \dots, S_r \leq s_r) &= \mathbb{P}(Z_{s_1} \geq 1, \dots, Z_{s_r} \geq r) \\ &= \mathbb{P}(X_{s_1} \geq 1, \dots, X_{s_r} \geq r) \\ &= \mathbb{P}(T_1 \leq s_1, \dots, T_r \leq s_r). \end{aligned}$$

Consequently, the S_k are finite almost surely and the differences $\eta_k = S_k - S_{k-1}$ are iid $\text{Exp}(\lambda)$ variables. Finally, almost surely, simultaneously for all $t \geq 0$,

$$Z_t = \sum_{k=1}^{\infty} \mathbf{1}\{Z_t \geq k\} = \sum_{k=1}^{\infty} \mathbf{1}\{S_k \leq t\}.$$

This is the exponential waiting-time construction, so Z is a Poisson process of rate λ . $\square$

11.1.3 Transition semigroups and generators

The family $\{\mathbf{P}_t : t \geq 0\}$ is called the *transition semigroup* of the chain.

Theorem 11.1.2. *The family $\{\mathbf{P}_t : t \geq 0\}$ is a stochastic semigroup; that is, it satisfies the following:*

- (a) $\mathbf{P}_0 = \mathbf{I}$, the identity matrix;
- (b) $\mathbf{P}_t$ is stochastic; that is, $\mathbf{P}_t$ has non-negative entries and row sums equal to 1;
- (c) the Chapman–Kolmogorov equations, $\mathbf{P}_{s+t} = \mathbf{P}_s \mathbf{P}_t$ if $s, t \geq 0$.

Proof. Part (a) is immediate.

For part (b), let $\mathbf{1}$ be the column vector of ones. Then

$$(\mathbf{P}_t \mathbf{1})_i = \sum_{j \in S} p_{ij}(t) = \mathbb{P} \left(\bigcup_{j \in S} \{X(t) = j\} \mid X(0) = i \right) = 1.$$

For part (c), using the Markov property and time homogeneity,

$$\begin{aligned} p_{ij}(s+t) &= \mathbb{P}(X(s+t) = j \mid X(0) = i) \\ &= \sum_{k \in S} \mathbb{P}(X(s+t) = j \mid X(s) = k, X(0) = i) \mathbb{P}(X(s) = k \mid X(0) = i) \\ &= \sum_{k \in S} p_{ik}(s) p_{kj}(t). \end{aligned}$$

Therefore, $\mathbf{P}_{s+t} = \mathbf{P}_s \mathbf{P}_t$. $\square$

Let $\tilde{\mathbf{P}}$ be the transition matrix of a discrete-time Markov chain. The family $\{\tilde{\mathbf{P}}^n : n = 0, 1, \dots\}$ can also be seen as a stochastic semigroup and it also satisfies the Chapman–Kolmogorov equations. The matrix $\mathbf{P}_t$ plays the role of $\tilde{\mathbf{P}}^n$, but now we have intermediate matrices between $\mathbf{P}_n$ and $\mathbf{P}_{n+1}$, which was not the case for discrete-time Markov chains.

Definition 11.1.3. The semigroup $\{\mathbf{P}_t : t \geq 0\}$ is called *standard* if $\mathbf{P}_t \rightarrow \mathbf{I}$ as $t \downarrow 0$, which is to say that $p_{ii}(t) \rightarrow 1$ and $p_{ij}(t) \rightarrow 0$ for $i \neq j$ as $t \downarrow 0$.

Exercise 11.1.3. Find a stochastic semigroup that is not standard.

Definition 11.1.4. If the following right limit exists,

$$\lim_{h \downarrow 0} \frac{1}{h} (\mathbf{P}_h - \mathbf{I}) = \mathbf{G},$$

the matrix $\mathbf{G}$ is called the *generator* of the semigroup.

Theorem 11.1.4. Suppose that S is finite. If the stochastic semigroup $\{\mathbf{P}_t : t \geq 0\}$ is standard, then its generator $\mathbf{G}$ exists and

$$\mathbf{P}_t = e^{t\mathbf{G}} \quad \text{for every } t \geq 0.$$

Proof. This proof can be found in [EN99].

Work with the induced maximum-row-sum norm. For $\delta > 0$, set

$$C_\delta := \int_0^\delta \mathbf{P}_s ds.$$

Standardness gives

$$\frac{1}{\delta} C_\delta = \int_0^1 \mathbf{P}_{\delta u} du \longrightarrow \mathbf{I} \quad \text{as } \delta \downarrow 0.$$

Hence C_δ is invertible for every sufficiently small $\delta > 0$. Fix one such δ .

The semigroup property gives, for $h > 0$,

$$\begin{aligned} (\mathbf{P}_h - \mathbf{I})C_\delta &= \int_0^\delta (\mathbf{P}_h \mathbf{P}_s - \mathbf{P}_s) ds \\ &= \int_0^\delta (\mathbf{P}_{h+s} - \mathbf{P}_s) ds \\ &= \int_\delta^{\delta+h} \mathbf{P}_u du - \int_0^h \mathbf{P}_u du. \end{aligned}$$

Therefore,

$$\frac{\mathbf{P}_h - \mathbf{I}}{h} = \left(\frac{1}{h} \int_\delta^{\delta+h} \mathbf{P}_u du - \frac{1}{h} \int_0^h \mathbf{P}_u du \right) C_\delta^{-1}.$$

The semigroup is right-continuous at every $t \geq 0$, since

$$\mathbf{P}_{t+h} - \mathbf{P}_t = \mathbf{P}_t(\mathbf{P}_h - \mathbf{I}) \longrightarrow \mathbf{0} \quad \text{as } h \downarrow 0.$$

Consequently,

$$\lim_{h \downarrow 0} \frac{\mathbf{P}_h - \mathbf{I}}{h} = (\mathbf{P}_\delta - \mathbf{I})C_\delta^{-1} =: \mathbf{G}.$$

Thus the generator exists.

For $t \geq 0$, the right derivative is

$$\lim_{h \downarrow 0} \frac{\mathbf{P}_{t+h} - \mathbf{P}_t}{h} = \mathbf{P}_t \mathbf{G}.$$

If $t > 0$, then

$$\begin{aligned} \frac{\mathbf{P}_t - \mathbf{P}_{t-h}}{h} &= \mathbf{P}_{t-h} \frac{\mathbf{P}_h - \mathbf{I}}{h} \\ &= \mathbf{P}_t \frac{\mathbf{P}_h - \mathbf{I}}{h} + (\mathbf{P}_{t-h} - \mathbf{P}_t) \frac{\mathbf{P}_h - \mathbf{I}}{h} \\ &\longrightarrow \mathbf{P}_t \mathbf{G}. \end{aligned}$$

To justify the second limit, observe that

$$\|\mathbf{P}_t - \mathbf{P}_{t-h}\|_\infty = \|\mathbf{P}_{t-h}(\mathbf{P}_h - \mathbf{I})\|_\infty \leq \|\mathbf{P}_h - \mathbf{I}\|_\infty \longrightarrow 0,$$

because stochastic matrices have maximum-row-sum norm 1. Moreover, $\mathbf{P}_t$ commutes with every difference quotient $(\mathbf{P}_h - \mathbf{I})/h$, and hence with $\mathbf{G}$. Therefore,

$$\frac{d}{dt} \mathbf{P}_t = \mathbf{G} \mathbf{P}_t, \quad \mathbf{P}_0 = \mathbf{I}.$$

Corollary 10.1.1 yields $\mathbf{P}_t = e^{t\mathbf{G}}$ for every $t \geq 0$. □

References

- [EN99] Klaus-Jochen Engel and Rainer Nagel. *One-parameter semigroups for linear evolution equations*. en. 2000th ed. Graduate Texts in Mathematics. New York, NY: Springer, Oct. 1999.